

LC 2050 - Parallel Courses III

(Amazon, Stripe)

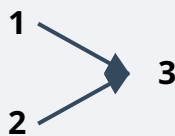
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Concepts & Observation

In **PC-I** all courses took time of 1 month hence one BFS level finished together.

In **PC-III** each course takes different time to finish. *So BFS level dont work anymore.*



1 : 3
2 : 2
3 : 5

then even if 2 finished, 1 still running.

so:

- cant start 3 untill 3 months have passed
- 3 cant start after 2 month because 1 is still running

So total = 3 + 5 → 8

hence

$$\text{finish}[v] = \max(\text{finish}[v], \text{finish}[u] + \text{time}[v])$$

↔ DP relation

Kahn is also required because a node becomes ready only after all prereq are done.

C++ Implementation

```

class Solution {
public:
    int minimumTime(int n, vector<vector<int>>& relations, vector<int>& time) {
        vector<vector<int>> adj(n+1);
        vector<int> indeg(n+1);
        for(auto &it : relations){
            int u = it[0];
            int v = it[1];
            adj[u].push_back(v);
            indeg[v]++;
        }

        vector<int> earliest(n+1, 0);
        //to store ek course node earliest kya time me khtm ho skta

        queue<int> q;
        for(int i = 1; i <= n; i++){
            if(indeg[i] == 0){
                q.push(i);
                earliest[i] = time[i-1];

                //-1 cause its 0 indexed
                //and source nodes apne time pe hi finish honge
                //koi parent nahi jinka max time consider krna pade
            }
        }

        while(!q.empty()){
            int curr = q.front();
            q.pop();

            for(auto &it : adj[curr]){
                earliest[it] = max(earliest[it], earliest[curr] + time[it-1]);
                //slowest parent decides time

                indeg[it]--;
                if(indeg[it] == 0){
                    q.push(it);
                }
            }
        }

        int ans = 0;
        for(int i = 1; i <= n; i++){
            ans = max(ans, earliest[i]);
        }
        return ans;
    }
};

```

```
}  
};
```